

Periodic Table

The Periodic Table arranges the elements according to their chemical properties.

Metals are found to the left hand side of the zig-zag on the table. **Non-metals** are found to the right.

Elements on either side of the zig-zag are called **semi-metals** or **metalloids**, showing characteristics of both metals and non-metals.

Hydrogen is a non-metal that behaves like a Group I element sometimes and a Group VII at other times. It is sometimes shown by itself in the table.

All metals with the same chemical properties are arranged in **groups** (vertical columns). They are numbered from I - VIII across the table.

The horizontal rows are called **periods**.

I		II																III	IV	V	VI	VII	VIII
1 H 1.008																			2 He 4.003				
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18						
11 Na 22.99	12 Mg 24.30	Transition metals										13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95						
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80						
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3						
55 Cs 132.9	56 Ba 137.3	57 *La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)						
87 Fr (223)	88 Ra (226)	89 §Ac (227)	104 Unq (261)	105 Unp (262)	106 Unh (263)	107 Uns (262)	108 Uno (265)	109 Une (266)	110	111													
* Lanthanide series		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.2	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0								
§ Actinide series		90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)								
6 C 12.01	← Atomic number ← Symbol ← Relative atomic mass (atomic weight)																						
A relative atomic mass in brackets is the mass number of the isotope with the longest half-life																							

Trends Across the Table

Consider the row 3 elements.

- Go from metals → semi-metals → non-metals.
- Reactivity (ability to react) decreases.

- Conductivity decreases.
i.e. Metals conduct electricity well - non-metals are insulators.
- Melting and boiling points increase from Group I - IV, then decrease from Group V - VIII.

The stronger the bonding between particles, the higher the melting and boiling point.

e.g. H_2O melts at $0\text{ }^\circ\text{C}$, NaCl melts at $801\text{ }^\circ\text{C}$. NaCl has the stronger force between its particles.

Trends Down a Group

- Reactivity increases down a metal group (I-III) - decreases down a non-metal group (IV-VIII).
e.g. Fr (francium) is the most reactive metal (bottom LHS of the table).
F is the most reactive non-metal (top RHS).
- Melting and boiling point decreases down metallic groups (I-III) - increases down non-metal groups.

Special Names of Elements

Alkali Metals

- Group I
- React with H_2O to form alkaline or basic solutions.

Alkaline Earth Metals

- Group II.

Halogens

- Group VII.

Inert (Noble) Gases

- Group VIII.
- Unreactive.
- Exist as single atoms.

Transition Metals

- Found in the middle of the table.
- Elements near each other have similar properties.
- Tend to form coloured compounds with non-metals.